

Project Title

Resume - Job Matching & Talent-Marketplace Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

This project presents the design and implementation of a Resume–Job Matching and Talent-Marketplace Engine aimed at optimizing the recruitment pipeline. Modern hiring platforms must match large pools of résumés against job postings, normalize inconsistent skill vocabularies, assign candidates to roles optimally, and assemble minimal skill sets to staff teams. To address this, the engine ingests and indexes textual data, utilizing advanced string algorithms to parse and score candidate-to-job fit. It employs network flow algorithms, such as Bipartite Matching, to compute the optimal assignment of candidates to available roles, ensuring maximum placement efficiency. Finally, it applies approximation techniques to solve the NP-hard problem of reporting the minimum skill set required to staff a given team. By hand-implementing these core algorithms, the system provides a scalable, computationally efficient solution to complex human resource allocation challenges.

Signature of the Guide